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CONVENTIONAL VALVE OPERATIONS

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The impact of annuloplasty ring or band implantation on post-repair mitral valve haemodynamic performance

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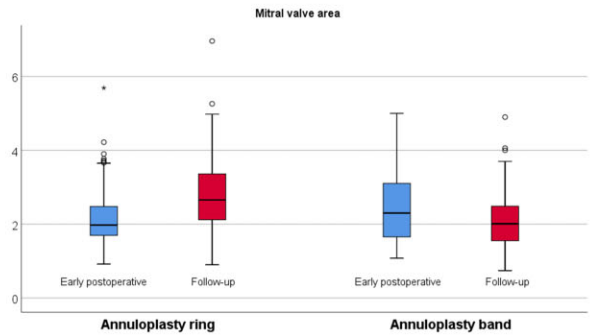
Post-repair mitral valve hemodynamic performance: does the type of annuloplasty device matter?

Summary

Population: Patients undergoing mitral valve repair for degenerative disease.
Intervention: Valve repair with semi-rigid annuloplasty ring (n=364) or partial band (n=171).
Comparison: Diastolic mitral valve hemodynamic performance, assessed by mitral valve gradient and valve area.
Outcome: Band did not offer a durable improvement in diastolic valve performance.



At **follow-up**, no differences in mitral valve area (coefficient -0.086 cm2, SE 0.120, P=1.00) was found on multivariable analysis.

Legend: SE: standard error.

Abstract

OBJECTIVES: The clinical importance of optimal post-repair mitral valve diastolic performance is increasingly being recognized. The haemodynamic effect of a partial annuloplasty band implantation, in comparison to a full ring, remains insufficiently explored.

METHODS: Patients undergoing mitral valve repair for pure degenerative disease between 2011 and 2019 at 2 experienced heart valve centres were eligible for inclusion. Exclusion criteria were concomitant procedures other than tricuspid valve repair and ablation

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procedures for atrial fibrillation. Pre-discharge and follow-up echocardiograms (1–4 years after surgery) were analysed to assess haemodynamic mitral valve performance.

RESULTS: Of 535 patients meeting the inclusion criteria, 364 (68.0%) patients underwent full annuloplasty ring and 171 (31.0%) partial band implantation. On predischarge echocardiogram, post-repair mitral valve gradient and area did not differ between groups [2.89 mmHg (IQR 2.26–3.72) vs 2.60 mmHg (IQR 1.91–3.55), $P = 0.19$ and 1.98 cm^2 (IQR 1.66–2.46) vs 2.03 cm^2 (IQR 1.55–3.06), $P = 0.15$]. However, multivariable linear regression analysis demonstrated band annuloplasty as a determinant of larger valve area (coefficient 0.467 cm^2, standard error 0.105, $P < 0.001$). On multivariable analysis, no significant impact on post-repair gradient was observed (−0.370 mmHg, standard error 0.167, $P = 0.36$). At follow-up, the differences between groups disappeared and multivariable regression analysis failed to demonstrate a significant impact of annuloplasty device type on mitral valve gradient (coefficient −0.095 mmHg, standard error 0.171, $P = 1.00$) or area (coefficient −0.085 cm^2, standard error 0.120, $P = 1.00$). These results were confirmed with a linear mixed model analysis.

CONCLUSIONS: Partial band annuloplasty was related to an improved haemodynamic profile directly after valve repair for degenerative disease but the effect was short-lived. Our results suggest that the type of annuloplasty device has no durable impact on diastolic valve performance.

Keywords: Mitral valve repair • Annuloplasty • Mitral valve haemodynamics

ABBREVIATIONS

CI	Confidence interval
IQR	Interquartile range
SD	Standard deviation

INTRODUCTION

Surgical mitral valve repair is the gold standard treatment for degenerative valve disease with excellent clinical outcomes [1]. With developments in the field and refinements in surgical technique occurring over the last decades, valve repair durability has constantly improved. In dedicated centres, high repair rates have been achieved and recurrent regurgitation is infrequently seen after an initial good repair result [2, 3].

A number of controversies in the field of reconstructive mitral valve surgery persist to date. Optimal post-repair diastolic haemodynamic performance is important to prevent the occurrence of residual symptoms or impaired exercise capacity [4–7]. Post-repair haemodynamics might be related to the technical aspects of valve repair surgery, among others to the type of annuloplasty device used [5, 8]. Annular remodelling and stabilization remain the cornerstone of surgical valve repair durability. Full semi-rigid rings and partial flexible bands are the most common devices used. The preference for the type of device remains subjective as the number of comparative studies on the results of valve repair with different types of annuloplasty devices remains extremely limited. Nevertheless, it has been suggested that partial flexible bands result in superior post-repair diastolic haemodynamic performance in patients undergoing valve repair for degenerative disease [5, 9].

The aim of our study was to assess the post-repair haemodynamic mitral valve performance in a cohort of patients who underwent surgical valve repair for degenerative disease. We aim to compare the effect of the type of annuloplasty device used at the time of surgery on post-repair haemodynamic performance shortly after surgery and at follow-up.

METHODS

Patients

Patients ≥18 years of age undergoing surgical intervention at the Department of Cardiovascular Surgery at Hospital Clínic de Barcelona, Barcelona, Spain, or Department of Cardiothoracic Surgery at Leiden University Medical Center, Leiden, Netherlands, between January 2011 and December 2019 were eligible for inclusion. Only patients with severe mitral valve regurgitation caused by degenerative valve disease were included. Exclusion criteria were other types of mitral valve disease and concomitant surgery other than tricuspid valve repair and/or concomitant ablation for atrial fibrillation. Patients who did not undergo an annuloplasty band or ring implantation (e.g. due to the presence of severe mitral annular calcification) were also excluded.

Ethics statement

The local institutional ethics committees of the Hospital Clínic de Barcelona, Barcelona, Spain (number HCB/2021/0248, date of approval 3 February 2021; patient consent was waived) and Leiden University Medical Center, Leiden, Netherlands (number P16.003, date of approval 14 June 2021; written patient consent was obtained) approved this study.

Data collection

Preoperative, intraoperative and postoperative data were collected from our computerized patient databases. For patients in whom follow-up was not centralized, follow-up echocardiographic studies and reports were imported and studied.

Surgical technique and annuloplasty device characteristics

All operations were performed by experienced mitral valve surgeons. Standard cannulation techniques were used, adjusted to

the type of surgical approach (central canulation in case of median sternotomy and peripheral canulation in case of minimally invasive surgery). The type of leaflet repair and annuloplasty device implantation was left at the discretion of the operation surgeon. Anterior mitral valve leaflet prolapse was treated predominantly by implantation of artificial neochords. For posterior mitral valve leaflet prolapse, a combination of chordal replacement and leaflet resection techniques were used. Commissural prolapse was treated by commissural closure, chordal replacement or papillary muscle head repositioning.

Sizing of the annuloplasty device was based on the surface area of the anterior mitral valve leaflet. In case of annuloplasty ring implantation, no under- or oversizing was performed. The rings and bands were implanted in a standard fashion, using interrupted mattress sutures. In case of annuloplasty band implantation, care was taken that the tips of the band were implanted at the height of both trigones. This technique was predominantly used in Barcelona.

To correct for the differences in annuloplasty rings and bands design and geometric characteristics, the posterior part of all annuloplasty devices was measured manually (the part corresponding to the intertrigonal length of the posterior annular perimeter) for all devices used. The results of the measurements are presented in Supplementary Material, Fig. S1.

Study endpoints

Primary study endpoints were post-repair mean resting mitral valve gradient and mitral valve area at (i) hospital discharge and at (ii) follow-up (1–4 years after the index operation). Secondary endpoint was a composite endpoint of freedom from mitral valve reintervention or all-cause mortality at 4 years after surgery.

Echocardiographic data collection

Mitral regurgitation severity was graded according to current recommendations using a multi-parametric approach, and was classified as mild (grade 1+), moderate (grade 2+), moderate–severe (grade 3+) and severe (grade 4+) [10]. Measurements of mitral valve gradients were acquired from continuous wave Doppler acquisitions of the diastolic inflow of the mitral valve. Mean mitral valve gradient was calculated as the average of 3 and 5 cycles or patients in sinus rhythm and atrial fibrillation, respectively. Left ventricular outflow tract and mitral valve velocity-time integrals were measured by the pulsed-wave Doppler method. Stroke volume was calculated with the use of the velocity-time integral of the left ventricular outflow tract and left ventricular outflow tract area. The continuity equation was used to calculate the post-repair mitral valve area.

Statistical analysis

Continuous data are presented as means and standard deviation (SD) for normally distributed data or medians and interquartile range (IQR) when not normally distributed. Categorical data are presented as counts and percentages. The distribution of variables was evaluated using the Kolmogorov–Smirnov test. Intergroup comparison of categorical variables was made using the χ^2 test and Fisher's exact test (when the expected value in any of the cells in the contingency table was <5). For continuous data, an independent 2-tailed Student's *t*-test or Mann–Whitney *U*-test was used if data were normally or non-normally distributed, respectively.

The correlation between annuloplasty band or ring size and post-repair mitral valve gradient or area was assessed using the Pearson's correlation coefficient method. Univariable and multivariable linear regression models, with post-repair mitral gradient or mitral valve area as dependent continuous variables at discharge and follow-up, were built to explore the factors associated with variable of interest. Variable inclusion was based on the previously established or presumed effect on mitral valve gradient and area. Variables included in the model were patient age, sex, blood haemoglobin value (only available at predischarge), heart rate, stroke volume, body surface area, type of posterior mitral valve leaflet repair (triangular resection or quadrangular resection with sliding), indentation closure, commissural closure, posterior annular perimeter length (according to the type and labelled device size), mitral regurgitation grade and type of annuloplasty device used (full ring or partial band). To assess for the presence of violations in model assumptions, residuals were plotted versus fitted values and investigated graphically. In multivariable models, the Bonferroni correction for multiple comparisons was used to calculate respective *P*-values.

To compare the clinical outcomes between groups, a survival analysis with a composite endpoint of mitral valve reintervention or all-cause mortality was performed. The Kaplan–Meier method was used to perform the time-to-event analysis and the differences between groups were compared with the log-rank test.

Additionally, linear mixed model analyses with mitral valve gradient or area as a dependent variable was performed. In the model, the intercept was added as a random effect. Follow-up time (in years) and type of annuloplasty device (full ring vs band) were added as fixed effect including an interaction term between these variables. Furthermore, fixed effects for the same covariates as in the previous regression analyses were included, namely age, sex, heart rate, stroke volume, body surface area, type of posterior mitral valve leaflet repair (triangular resection or quadrangular resection with sliding), indentation closure, commissural closure, posterior annular perimeter length (according to the type and labelled device size) and mitral regurgitation grade. Note that due the inclusion of a random intercept, the coefficients of the mixed model analyses do not have a marginal (i.e. group-level) meaning and should be interpreted as subject-specific.

All tests were 2-tailed and a *P*-value of <0.05 was considered statistically significant. Statistical analysis was performed using IBM Statistics for Windows, version 25.0 (IBM Corp. IBM SPSS Statistics for Windows, Version 2.0. Armonk, NY, USA) and R (R Foundation for Statistical Computing, Vienna, Austria).

RESULTS

Baseline characteristics and intraoperative details

In total, 590 patients were screened. We excluded 48 patients who did not undergo valve repair with the Cosgrove annuloplasty band (Edwards Lifesciences, Irvine, CA, USA), the Physio II ring (Edwards Lifesciences, Irvine, CA, USA) or the Memo 3D (Corcym, London, UK) ring, as the groups of patients undergoing valve repair with other types of annuloplasty devices were small and therefore considered not representative. Additionally, 7 patients who underwent edge-to-edge repair were excluded. The

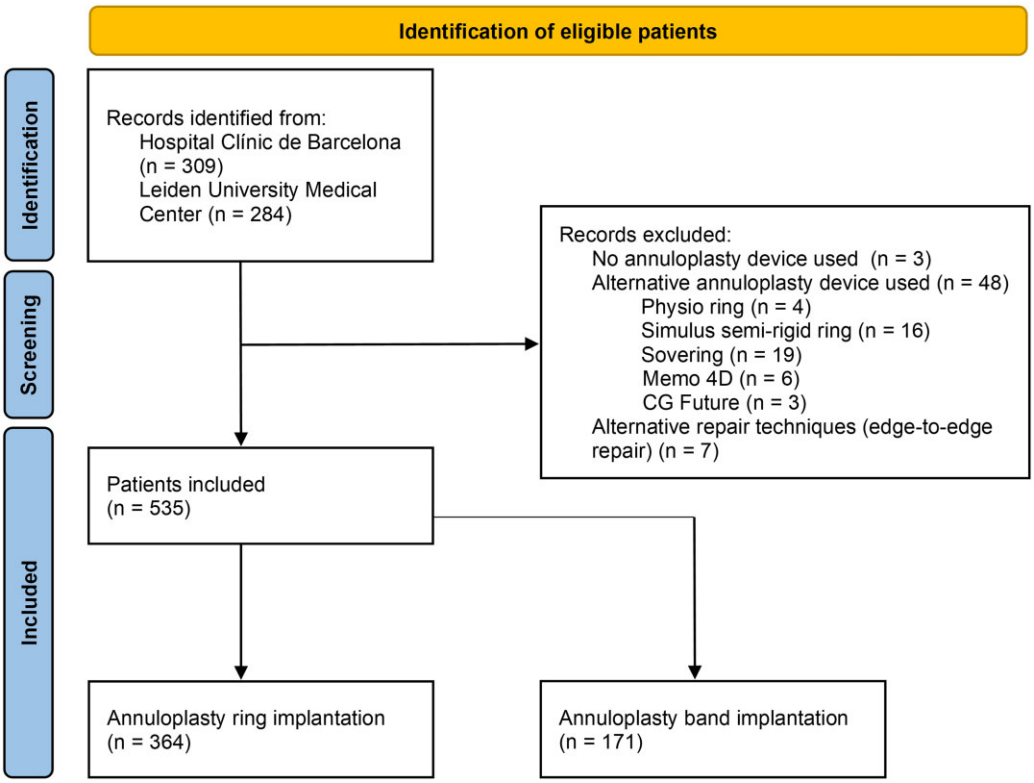

Figure 1: Study flow chart. After applying the study exclusion criteria, 535 patients were included.

final study cohort consisted of 535 patients (Fig. 1). Ring annuloplasty was performed in 364 (68.0%) patients while 171 (32.0) underwent partial band implantation. Of patients undergoing full ring implantation, 317/364 (87.1%) underwent Physio II and 47/364 (12.9%) Memo 3D implantation.

Baseline patient characteristics are presented in Table 1 and were comparable between groups. Patients undergoing annuloplasty ring implantation, however, presented with larger body surface area (1.90 m^2, SD 0.21, vs 1.83 m^2, SD 0.20, $P < 0.001$).

Repair of the posterior mitral valve leaflet was performed in the majority of patients from both groups (Table 2). Of notice, indentation closure was performed more often in the ring annuloplasty group (20.1% vs 12.9%, $P = 0.042$), while the frequency of commissural closure did not differ between groups. Larger annuloplasty devices were used in the ring group. Only 36/535 (6.7%) patients underwent an annuloplasty device implantation of size smaller than 30.

Early postoperative echocardiographic results

On predischarge echocardiogram (Table 3), mean resting mitral valve gradient did not differ between the ring and band annuloplasty groups (2.89 mmHg, IQR 2.26–3.72 vs 2.60 mmHg, IQR 1.91–3.55, $P = 0.19$). The mean mitral valve gradients according to the type of device used and labelled device size are presented in Fig. 2. Annuloplasty band demonstrated lower gradients particularly in patients with smallest annuloplasty device sizes (size 26 and 28). No difference in the post-repair mitral valve area was seen between groups (Table 3).

Pearson's correlation test revealed a statistically significant correlation between mean mitral valve gradient and posterior annular perimeter only in the ring annuloplasty group (Fig. 3). Accordingly, the correlation between post-repair mitral valve area and posterior annular perimeter was only significant in the ring annuloplasty group. However, the correlation was considered weak in both cases ($r = -0.337$ and $r = 0.210$, respectively).

Multivariable regression analyses demonstrated that the use of band annuloplasty was not related to a significant difference in post-repair resting mitral valve gradient (coefficient −0.370 mmHg, standard error 0.167, $P = 0.36$; Supplementary Material, Table S1 and Fig. S2) but was related to a larger post-repair mitral valve area (coefficient 0.467 cm^2, standard error 0.105, $P < 0.001$; Supplementary Material, Table S2 and Fig. S3).

Clinical results at follow-up

At 4 years after surgery, the estimated freedom from mitral valve reintervention or all-cause mortality was comparable between the annuloplasty ring and band group (94.4%, 95% CI 92.0–96.8%, vs 93.5%, 95% CI 89.8–97.2%, respectively, $P = 0.67$; Supplementary Material, Fig. 4). There were no reinterventions for mitral valve stenosis in this time period.

Echocardiographic result at follow-up

The median echocardiographic follow-up was 2.0 (IQR 1.4–2.8) years and 2.7 (IQR 2.5–3.6) years for the ring and band group, respectively, and was 74.8% complete for the mitral valve gradient and 68.1% complete for the mitral valve area measurements. Mean resting mitral valve gradient was lower in the ring annuloplasty group (2.40 mmHg, IQR 1.87–3.35 vs 2.70 mmHg, IQR 2.03–3.60, $P = 0.008$). Moreover, there was a statistically significant difference in the post-repair mitral valve area, favouring the

Table 1: Preoperative clinical and echocardiographic characteristics

	Ring (*N* = 364)	Band (*N* = 171)	*P*-value
Age	63.6 (IQR 55.8–71.0)	63.1 (IQR 56.1–71.6)	0.95
Gender (female)	131 (36.2)	56 (32.7)	0.44
Body surface area (m^2)	1.90 (SD 0.21)	1.83 (SD 0.20)	<0.001
Peripheral vascular disease	2 (0.5)	4 (2.3)	0.086
Chronic lung disease	28 (7.7)	10 (5.8)	0.44
Diabetes mellitus	17 (4.7)	13 (7.6)	0.17
Atrial fibrillation	121 (33.2)	50 (29.2)	0.35
Left ventricular end systolic diameter (mm)	34 (IQR 30–39)	35 (IQR 30–39)	0.94
Left ventricular end diastolic diameter (mm)	56 (IQR 51–61)	58 (IQR 52–61)	0.46
Left ventricular ejection fraction (%, continuous)	63 (IQR 58–68)	60 (IQR 59–65)	0.13
Impaired left ventricular function (categorical)			0.003
Moderate	43 (11.9)	6 (3.6)	
Poor	0 (0)	1 (0.6)	

Data are presented as *N* (%), means (SD) or medians (IQR).

Table 2: Intraoperative details on valve repair

	Ring (*N* = 364)	Band (*N* = 171)	*P*-value
Annuloplasty device size			<0.001
26	5 (1.4)	2 (1.2)	
28	11 (3.0)	18 (10.5)	
30	40 (11.0)	28 (16.4)	
32	104 (28.6)	54 (31.6)	
34	81 (22.3)	47 (27.5)	
36	51 (14.0)	21 (12.3)	
38	50 (13.7)	0 (0)	
40	22 (6.0)	1 (0.6)	
Posterior annular perimeter	64 (IQR 60–68)	58 (IQR 54–60)	<0.001
Posterior mitral valve leaflet	300 (81.3)	124 (71.7)	
Implantation of neochords	215 (59.1)	47 (27.6)	<0.001
Triangular resection	101 (27.9)	62 (36.7)	0.041
Quadrangular resection with leaflet sliding	69 (19.1)	18 (10.7)	0.015
Indentation closure	73 (20.1)	22 (12.9)	0.042
Anterior mitral valve leaflet	77 (21.0)	22 (12.8)	0.021
Implantation of neochords			
Commissures			
Closure	29 (8.0)	15 (8.8)	0.75
Other (neochords, papillary muscle head repositioning)	15 (4.1)	0 (0)	0.004
Tricuspid valve repair	133 (36.5)	3 (1.8)	<0.001
Concomitant ablation	76 (20.9)	13 (7.6)	<0.001

Data are presented as *N* (%) or medians (IQR).

Table 3: Postoperative echocardiographic results and blood haemoglobin value

	Ring (*N* = 364)	Band (*N* = 171)	*P*-value
Mean resting mitral valve gradient (mmHg)	2.89 (IQR 2.26–3.72)	2.60 (IQR 1.91–3.55)	0.19
Heart rate (min^{-1})	79 (IQR 68–88)	82 (IQR 72–90)	0.011
Stroke volume (ml)	68 (IQR 55–82)	47 (IQR 39–60)	<0.001
Haemoglobin value (mmol/l)	6.6 (IQR 6.0–7.1)	6.8 (IQR 6.0–7.5)	0.036
Mitral valve area (cm^2)	1.98 (IQR 1.66–2.46)	2.03 (IQR 1.55–3.06)	0.15
Mitral regurgitation grade			0.040
Mild (grade 1+)	76 (20.9)	25 (14.6)	
Moderate (grade 2+)	14 (3.9)	2 (1.2)	

Data are presented as *N* (%) or medians (IQR).

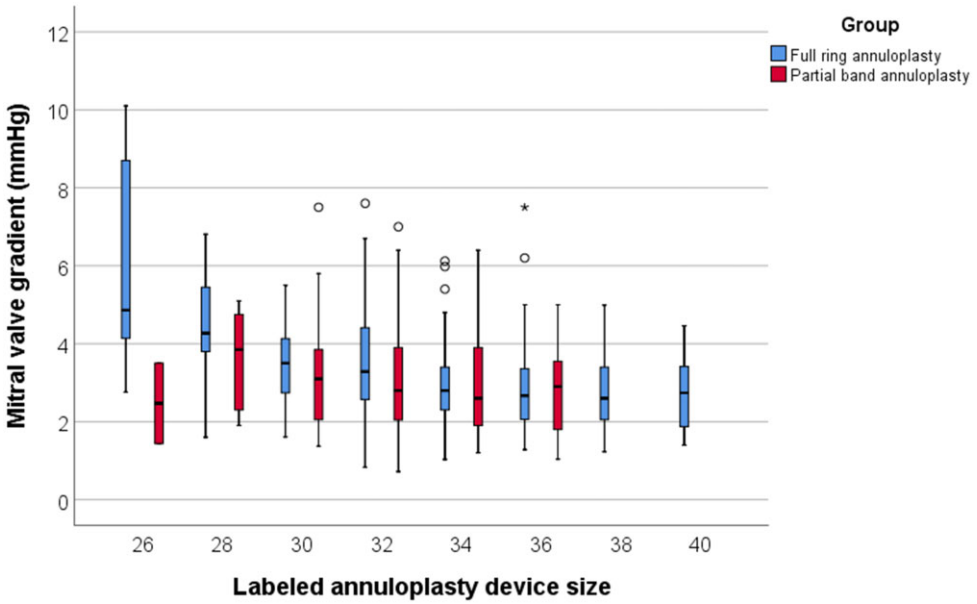

Figure 2: Graphical presentation of the mean mitral valve gradient according to the type of device used and labelled device size.

ring annuloplasty group (2.65 cm^2, IQR 1.96–3.34 vs 2.04 cm^2, IQR 1.51–2.63, $P < 0.001$). A comparison between the discharge and follow-up values is presented in Fig. 4.

Pearson's correlation test revealed a statistically significant correlation between mean mitral valve gradient and posterior annular perimeter only in the ring annuloplasty group. Accordingly, the correlation between post-repair mitral valve area and posterior annular perimeter was only significant in the ring annuloplasty group (Fig. 5). The correlation was considered moderate in case of mean mitral valve gradient ($r = -0.451$) and weak in case of mitral valve area ($r = 0.313$, respectively).

Contrary to the results seen at discharge, multivariable regression analyses failed to demonstrate a beneficial effect of partial band annuloplasty on post-repair mitral valve gradient (coefficient −0.095 mmHg, standard error 0.171, $P = 1.00$; Supplementary Material, Table S3 and Supplementary Material, Fig. S5) or mitral valve area (coefficient −0.085 cm^2, standard error 0.120, $P = 1.00$; Supplementary Material, Table S5 and Supplementary Material, Figure S6).

Linear mixed model analysis

At discharge, the mitral valve gradient was lower for a typical patient in the band annuloplasty group (coefficient −0.46 mmHg, 95% CI −0.75 to −0.17). Over time, the mean gradient decreased for a typical patient in the ring annuloplasty group (coefficient −0.10 mmHg per year, 95% CI −0.17 to −0.03). The interaction term between follow-up time and annuloplasty group indicated that the progression in mean gradient for a typical patient was 0.11 mmHg higher per year (95% CI 0.01–0.21) in the band annuloplasty group as compared to the ring annuloplasty group.

The mitral valve area was larger for a typical patient in the band annuloplasty group at discharge (coefficient 0.47 cm^2, 95% CI 0.01–0.94). Over time, the mitral valve area for a typical patient increased in the ring annuloplasty group (coefficient 0.27 cm^2 per year, 95% CI 0.19–0.35). The interaction term between follow-up time and annuloplasty group indicated that the progression in mitral valve area for a typical patient was 0.30 cm^2 lower per (95% CI −0.42 to −0.19) in the band annuloplasty group as compared to the ring annuloplasty group.

DISCUSSION

The results of our study demonstrate that while the use of an annuloplasty band might have a beneficial effect on post-repair mitral valve area and gradient shortly after surgery, this effect is likely short-lived. At follow-up, diastolic post-repair mitral valve performance was unrelated to the use of either a band of ring. Our results provide novel insights into the post-repair diastolic mitral valve performance and the longitudinal changes that occur over the years after surgical repair.

To the best of our knowledge, the evolution of mitral valve area after mitral valve repair has not yet been assessed in a manner comparable to our study. On the other hand, the evolution of mitral valve gradients in our study is comparable to previously published results from single-arm studies. We have previously shown that after valve repair with a full ring, a gradual decrease in mitral valve gradient (years after surgery) can be expected [4]. The results of the current study are in line with our previous observations. Moreover, Murashita *et al.* have studied the evolution of mitral valve gradient in patients having undergone mitral valve repair with a partial band [11]. When compared to the intraoperative gradient, the authors reported higher gradients at early and mid-term follow-up. Similar observations are seen in our study as the mitral valve gradient following band annuloplasty increased at mid-term follow-up when compared to pre-discharge measurements. The evolution of post-repair mitral valve gradients are therefore in line with results previously reported.

Implantation of an annuloplasty device of any type is bound to result in a reduction of maximum valve area and a fixation of the posterior annular perimeter length [12]. In healthy individuals, the posterior part of the mitral valve annulus will lengthen during cardiac cycle, in particular at the annular area corresponding to indentations, normally present between scallops of the posterior

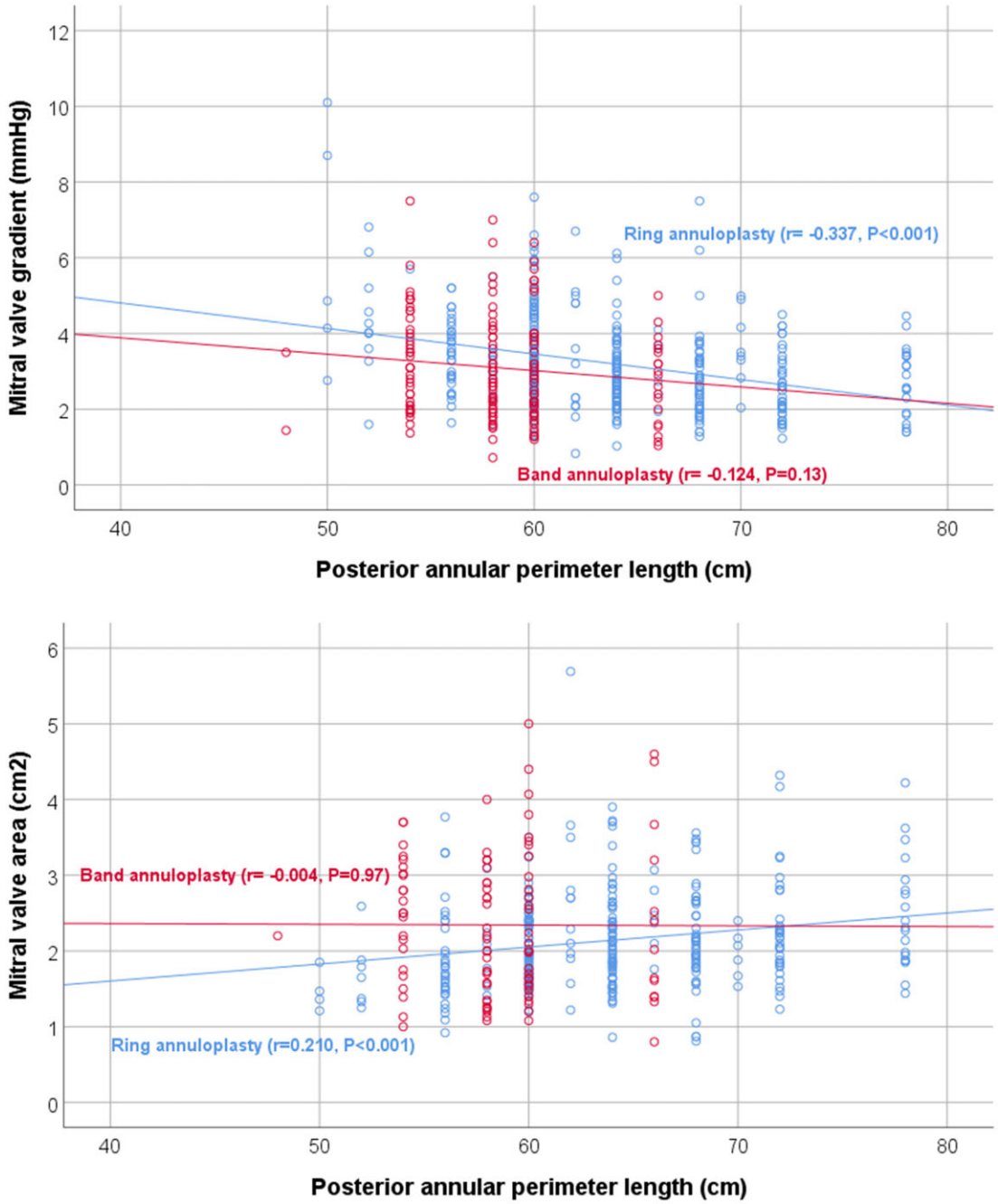

Figure 3: Graphical presentation of the relation between mean mitral valve gradient (above) and mitral valve area (below) and posterior annular perimeter length shortly after the index operation. While a correlation between the posterior annular perimeter and mitral valve area was clear for the annuloplasty ring group, this was not the case in the band annuloplasty group.

leaflet. During exercise, the posterior annulus will accommodate an additional increase in the mitral valve area by further expanding during diastole. Following implantation of any prosthetic annuloplasty device, the length of the posterior annulus will remain fixed in a systolic configuration, preventing any augmentation during diastole [13]. Transvalvular gradients higher than normally seen in healthy individuals are therefore inevitable in any type of mitral valve repair when annular remodelling and stabilization is performed, due to the corresponding decrease in maximal mitral valve area.

Somehow surprisingly, the mitral valve area following repair with an annuloplasty ring demonstrated an increase at mid-term follow-up when compared to predischarge values. The only meaningful explanation for our finding is that the mitral valve area in these patients is determined by factors other than the ring itself. A similar phenomenon has actually already been proven in patients with functional mitral regurgitation. Following valve repair by restrictive mitral annuloplasty, Bertrand *et al.* have demonstrated that the opening angle of the anterior leaflet (typically tethered in these patients) is more strongly correlated

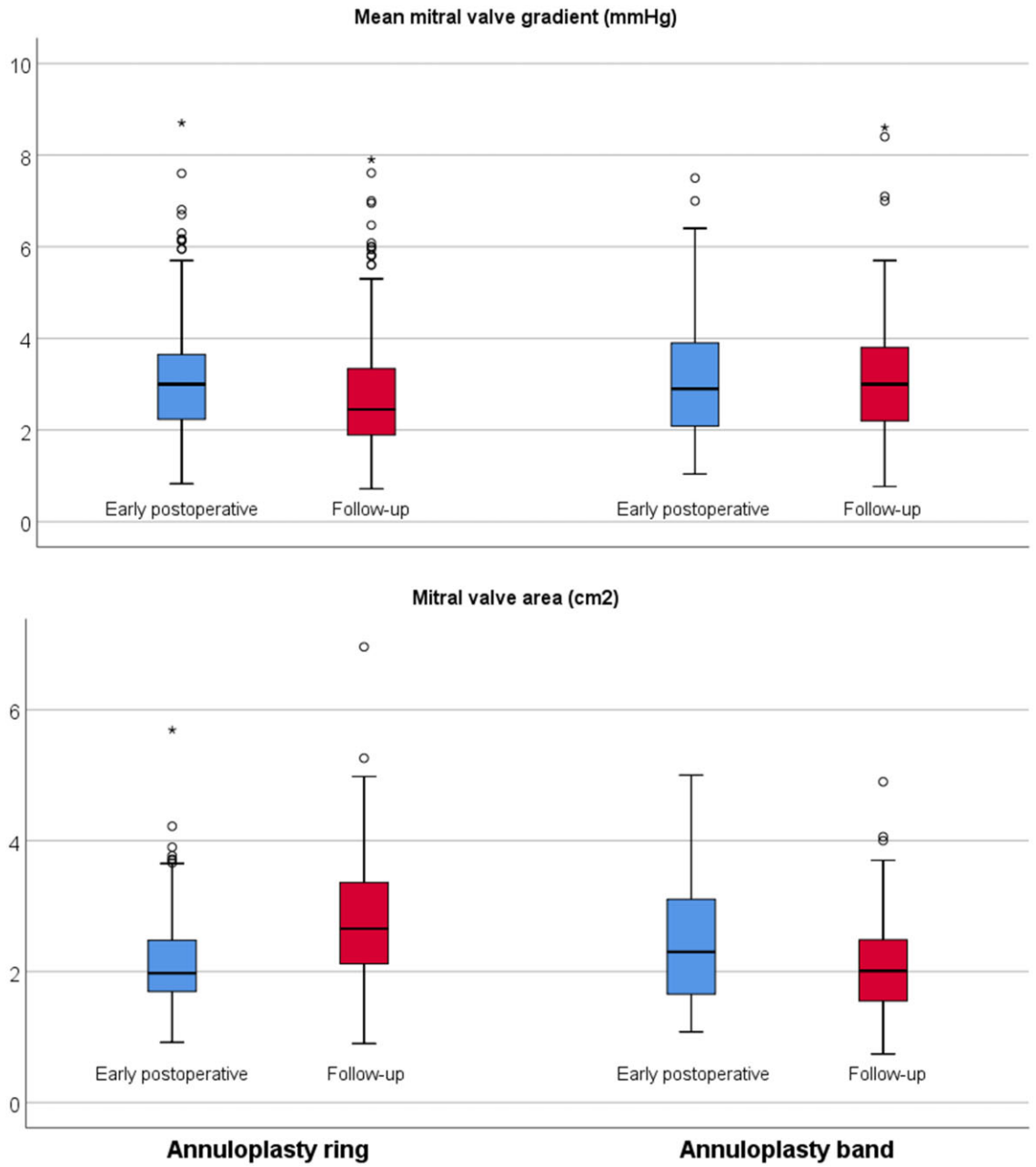

Figure 4: A comparison between the discharge and follow-up values for mitral valve gradient (above) and area (below) for the annuloplasty ring and band groups.

to the mitral valve gradient and effective orifice area than the size of the annuloplasty device [14]. Interestingly, it has been shown that in patients with ischaemic cardiomyopathy, left ventricular reverse remodelling is more pronounced after mitral valve repair with a ring than a band [15].

While the determinants of valve opening are clearly different in functional and degenerative mitral valve disease, reverse remodelling of the left ventricle is expected following successful resolution of mitral regurgitation and might result in differences in the opening properties of the mitral valve. Moreover, reverse remodelling is a treatment goal in surgical repair for degenerative mitral valve disease, aiming to reverse geometric changes caused by long-standing volume overload. While flexible annuloplasty devices have been shown to better preserve annular dynamics, they have also been linked to limited annular remodelling capacity [16]. The effect of these differences in degenerative mitral valve disease remains unstudied. Mitral valve opening is not an all or nothing process but adapts to haemodynamic and geometric characteristics of the left ventricle and the surrounding structures [17].

The evolution of mitral valve area in patients after valve repair with a partial band demonstrated a different pattern. In these patients, a decrease in post-repair gradients was not observed over time and the mitral valve area remained more or less stable. We can only speculate about the potential explanations for our findings; partial annuloplasty does not cover the anterior part of the mitral valve annulus and the opening properties of the anterior leaflet might be different after annuloplasty with a full ring or a partial band [12]. Moreover, endothelization and, to some extent, fibrosis and subsequent stiffening of the flexible band is

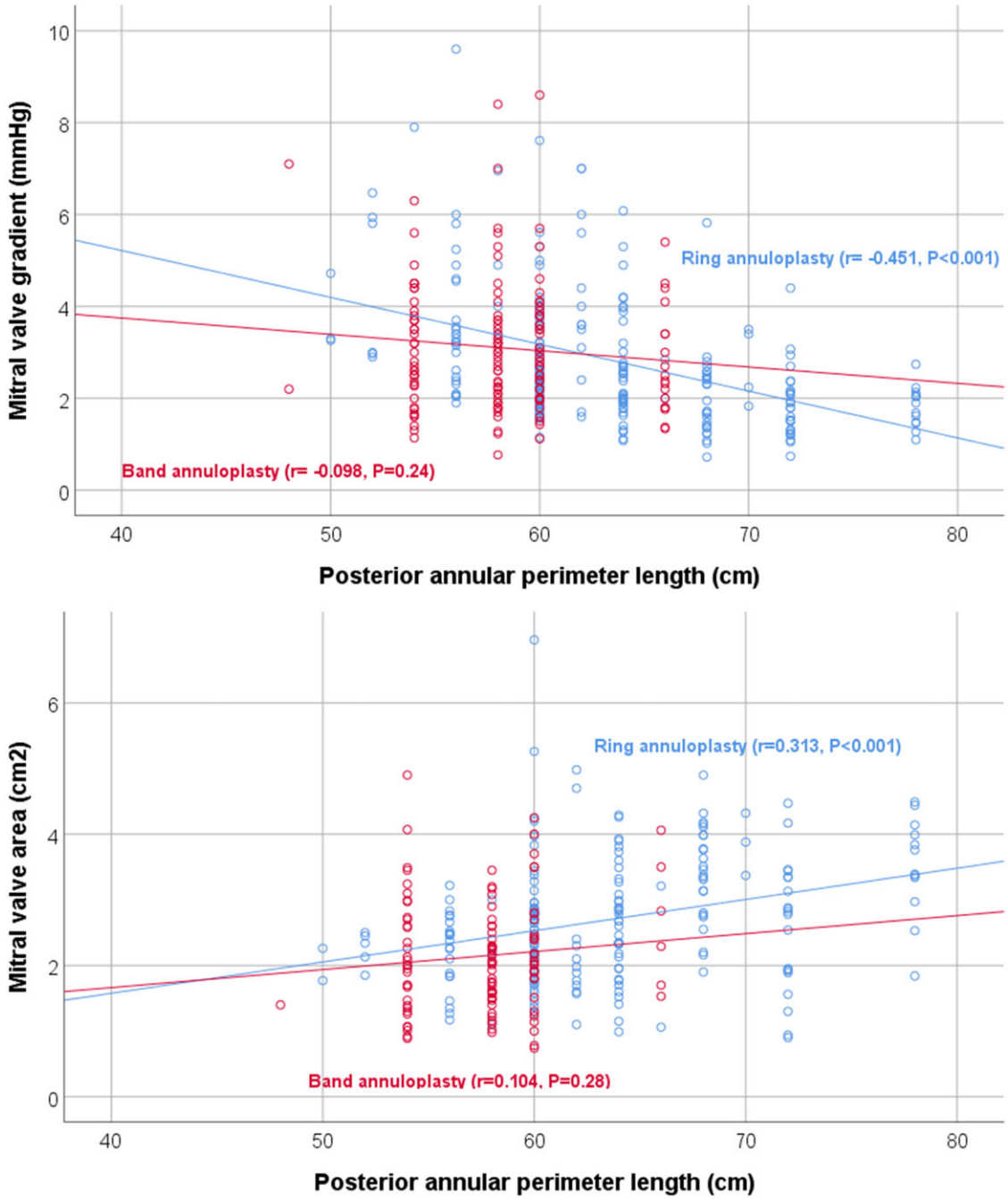

Figure 5: Graphical presentation of the relation between mean mitral valve gradient (above) and mitral valve area (below) and posterior annular perimeter length at mid-term follow-up.

bound to occur over time, eliminating the theoretical advantage of annuloplasty device flexibility.

We believe that the results of our study, together with the existing literature, confirm our hypothesis that there is no strong evidence supporting the superiority of any annuloplasty device on late outcomes of valve repair. Valve repair techniques differ in regard to the type of annuloplasty device used; a most appealing conclusion would be that surgeons should aim for consistency in their practice in order to obtain the best results. However, certain situations might demand us to adjust to patient- and valve-related characteristics. In particular, repair of anterior mitral valve leaflet prolapse remains challenging and there is some evidence suggesting that valve repair durability is better when full annuloplasty devices rather than partial ones are used [18]. Moreover, the lack of patients having undergone small size annuloplasty device implantation in our study needs to be acknowledged. When a small size annuloplasty device is needed at the time of valve repair, a partial band might be favoured, as suggested by the results of our study and reported by others [9, 19].

Limitations

This is a retrospective study with limitations inherent to the study design. As various experienced surgeons have performed the

procedures, we cannot fully exclude the subjective effect of the technical aspects of leaflet repair or annular device sizing on our results. However, as no repair technique demonstrated an effect on post-repair valve characteristics, we reason that this does not impair the validity of our results. The postoperative clinical and echocardiographic follow-up scheme varied amongst patients and we assessed diastolic valve performance at a distinct follow-up period to collect sufficient data. We reason that the time elapsed after surgery did not play a significant role on diastolic valve performance in the specified study follow-up timeframe. Finally, only echocardiograms at rest were available for analysis. The inclusion of echocardiograms at rest enabled us to include a large group of patients in whom data was readily available. Performance of stress echocardiogram would require a different study design and would not be feasible in such a large cohort.

CONCLUSIONS

Our results suggest that the choice between a full ring or partial band annuloplasty at the time of mitral valve repair has no durable impact on the haemodynamic performance of the mitral valve. Early haemodynamics seemed favourable with the use of a partial band but at follow-up no difference could be observed.

SUPPLEMENTARY MATERIAL

Supplementary material is available at EJCTS online.

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DATA AVAILABILITY

Data will be made available on reasonable request to the corresponding author.

Author contributions

Anton Tomšič: Data curation; Formal analysis; Methodology; Writing—original draft. **Elena Sandoval:** Conceptualization; Data curation; Writing—review and editing. **Maria C. Meucci:** Data curation; Writing—review and editing. **Takeru Nabeta:** Data curation; Writing—review and editing. **Manuel Castella:** Conceptualization; Supervision; Writing—review and editing. **Anna Muro:** Data curation; Writing—review and editing. **Robert J.M. Klautz:** Conceptualization; Supervision; Writing—review and editing. **Nina Ajmone Marsan:** Data curation; Methodology; Writing—review and editing. **Daniel Pereda:** Conceptualization; Methodology; Supervision; Writing—review and editing. **Meindert Palmen:** Conceptualization; Methodology; Supervision; Writing—original draft.

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